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Implementing a Stack in Python

Class Stack:

Def init_(self):

Initializes an empty list to hold stack elements

Self.items = []

Def is_empty(self):

Returns True if the stack is empty; otherwise,

False return len(self.items) == 0

Def push(self, item):

#Adds item to the end of the list (top of the stack)

Self.items.append(item)

Def pop(self):

Removes and returns the last item (top of the stack), if not empty

return self.items.pop() if not self.is_empty() else None

Def peek(self):

Returns the last item without removing it, if not empty

return self.items[-1] if not self.is_empty() else None

Def size(self):

Returns the number of items in the stack

Return len(self.items)

Implementing a Queue in Python

Class Queue:

Def init (self):

Initializes an empty list to hold queue elements

Self.items = []

Def is_empty(self):

#Returns True if the queue is empty; otherwise, False

Return len(self.items) == 0

Def enqueue(self, item):

#Adds an item to the end of the list (end of the queue)

Self.items.append(item)

Def dequeue(self):

Removes and returns the first item (front of the queue), if not empty

return self.items.pop(0) if not self.is_empty() else None

Def size(self):

Returns the number of items in the queue

Return len(self.items)